

Comparing Human and Machine Visual Saliency: A Comprehensive Review

Introduction

Understanding where humans look in images (human visual saliency) and how machine vision models attend to visual information is a key challenge at the intersection of computer vision and cognitive science. **Visual saliency** refers to the priority of certain image regions in capturing attention. Human saliency is typically measured by eye fixations during free viewing, whereas machine models can produce **saliency maps** or attention weights indicating the image parts most influential to their decisions. This review synthesizes state-of-the-art research on human saliency prediction and AI attention mechanisms, with a focus on two proposed research phases: (1) comparing human saliency models to various computer vision model attentions, and (2) using human saliency as a prior to improve machine vision models. We cover the latest saliency prediction models, attention mechanisms in CNNs and Transformers, evaluation metrics and benchmarks, findings on human-machine saliency alignment, and attempts to integrate human gaze data into model training. The goal is to assess the viability and novelty of aligning machine attention with human visual salience for better efficiency, performance, and interpretability.

Human Visual Saliency Prediction: State of the Art

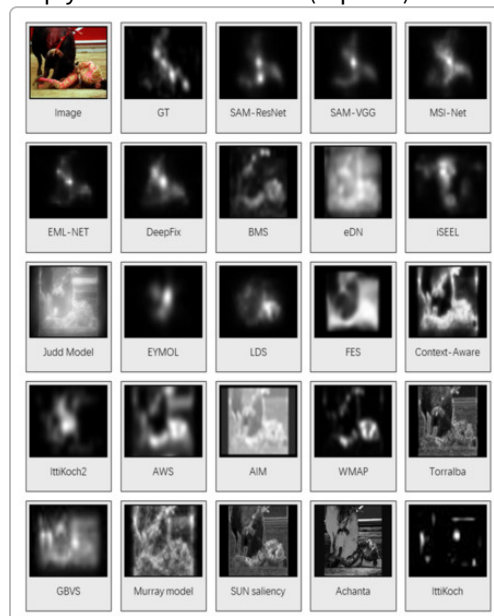
Early computational models of human saliency were inspired by neurobiology and psychology. They often split into **bottom-up** (stimulus-driven) and **top-down** (task-driven) approaches ¹ ². Classic bottom-up models like **Itti & Koch's model (1998)** used multi-scale feature contrasts (color, orientation, intensity) to simulate attention shifts ³. **Graph-Based Visual Saliency (GBVS)** by Harel (2007) introduced a Markovian approach to combine feature maps, enhancing conspicuity ⁴. Many variants leveraged contrast, information theory, and frequency domain analysis (e.g. Bruce & Tsotsos' s AIM model using information maximization ⁵, Hou' s spectral residual method ⁶) to predict which regions draw the eye. Top-down models incorporated scene context and task: e.g. Torralba' s Bayesian model combined scene priors with feature salience ⁷, and learned models (Judd et al., 2009) used machine learning (SVMs, regression) on combined features ⁸. These classical models, while insightful, plateaued in performance due to reliance on hand-crafted features ⁹.

Deep Learning Saliency Models: Around 2014, deep networks began to dominate saliency prediction. One early CNN-based model, eDN by Vig et al. (2014), showed the promise of learned features ¹⁰. Subsequent models dramatically improved accuracy by learning from large eye-tracking datasets. Notable examples include:

- **DeepGaze II** (Kümmerer et al., 2016) – used deep features from a pretrained VGG-19 network, fine-tuned on saliency data (SALICON and MIT datasets) ¹¹. It achieved top performance on several metrics (e.g. highest AUC-Borji = 0.86 on MIT benchmark) ¹².
- **SALICON** (Huang et al., 2015) – a model named after the SALICON dataset (obtained via crowdsourced mouse-tracking as a proxy for eye fixations). It performed strongly especially on metrics accounting for center-bias (e.g. best shuffled AUC = 0.74) ¹².

- **DeepFix** (Kruthiventi et al., 2017) – an early deep model incorporating location-biased convolutions to account for center-bias in fixations.
- **Saliency Attentive Models (SAM-VGG, SAM-ResNet)** (Cornia et al., 2018) – models that combined deep features with an attentive LSTM mechanism to refine saliency maps. These were top performers on MIT Saliency Benchmark, with SAM-ResNet achieving AUC-Judd ≈ 0.88 on CAT2000 dataset ¹³.
- **EML-NET** (2019) – an ensemble multi-level network that achieved best overall scores on MIT300 (winning the MIT Saliency Benchmark) ¹⁴.
- **SalGAN** (Pan et al., 2017) – a GAN-based saliency model producing smooth, human-like saliency predictions.

These deep models vastly outperform classic approaches. For instance, on the MIT300 benchmark (300 natural images with eye-tracking ground truth), all top 10 models are CNN-based ¹⁴. DeepGaze II, DeepFix, and EML-NET excel in AUC-based scores, while others like SAM and SalGAN score better on distribution-based metrics ¹⁵ ¹⁶. **Figure 1** illustrates a sample image's saliency map as predicted by several models versus human ground truth, showing that modern deep saliency models (bottom row) capture salient objects more sharply than classic models (top row)





Example saliency maps from different models on the same image (from CAT2000 dataset). Top row: classic models (Itti/Koch, GBVS, etc.) produce more diffuse or limited highlights. Bottom row: deep learning models (SAM-ResNet, SalGAN, EML-Net, etc.) focus sharply on the human-eye fixated regions (ground truth on left) ¹⁷ .

Beyond static images, there are dynamic saliency models for video (e.g. ACLNet, STRA-Net) which account for temporal attention; these likewise benefit from deep architectures and often use LSTM or 3D CNN components ¹⁸ ¹⁹ . The state of the art continues to advance, with recent models like **UNISAL**, **ML-Net**, and even **DeepGaze III** (which predicts entire scanpaths) pushing performance closer to the “infinite humans” upper bound (an idealized score if infinite observers were averaged) ²⁰ ²¹ . However, a performance gap remains between models and the human ground truth, highlighting that saliency prediction is not yet a solved problem ²² ²³ . Key challenges include modeling higher-level semantic factors (e.g. object meaning, task demands) that influence gaze ²⁴ .

Saliency Evaluation Metrics: To compare saliency maps against human data, researchers use a suite of complementary metrics ²⁵ :

- **AUC (Area Under ROC Curve)** and its variants: **AUC-Judd**, **AUC-Borji**, **sAUC**. These treat predicted saliency as a binary classifier on fixation vs. non-fixation points ²⁶ ²⁷ . For example, AUC-Judd considers saliency at fixated pixels as true positives and at all other pixels as false positives ²⁸ . AUC-Borji is similar but uses random non-fixations sampling ²⁹ . **Shuffled AUC (sAUC)** controls for center-bias by using other images’ fixations as negatives ²⁷ . Higher AUC indicates better fixation prediction (AUC ≈ 0.88 is top-tier ³⁰).
- **NSS (Normalized Scanpath Saliency):** The average normalized saliency at human fixation locations ³¹ . Essentially, it z-scores the model’s saliency map and computes the mean value at fixation coordinates. NSS > 1 indicates the model assigns significantly higher saliency to fixated points than to a random point ³² . This is a direct alignment measure (NSS is positively correlated with model performance; infinite-human NSS ~ 3.3 on MIT300) ³³ .
- **CC (Linear Correlation Coefficient):** Pearson’s correlation between the predicted saliency map and the ground-truth fixation density map ³⁴ . Ranges from -1 to 1; higher is better, treating saliency as continuous distribution.
- **SIM (Similarity or Intersection-over-Union):** Measures how similar the two saliency distributions are by summing the minimum value at each pixel after normalizing both to

probability distributions ³⁵. A SIM of 1 would mean the model's saliency distribution perfectly overlaps the human distribution ³⁶.

- **KL-Divergence:** A dissimilarity measure of how one distribution (model) diverges from the human distribution; lower KL indicates closer alignment ³⁷.
- **EMD (Earth Mover's Distance):** The minimal "work" needed to transform the model's saliency distribution into the human distribution, treating saliency as piles of sand to be moved ³⁸. Lower EMD means the spatial distribution of attention is closer to human.

Each metric captures different aspects (location hits, distribution shape, etc.), so saliency benchmarks report several metrics ²⁵. A robust saliency model should perform well across multiple metrics ³⁹.

Saliency and Attention in Computer Vision Models

Modern vision models incorporate or can produce their own form of "saliency" or attention maps, though these are not always directly comparable to human fixations. We consider three broad classes: **convolutional neural networks (CNNs)**, **vision transformers (ViTs)**, and **adaptive attention networks**, examining how each encodes visual attention.

- **CNNs and Implicit Saliency:** Standard CNNs for image recognition (e.g. ResNet, VGG) do not have an explicit attention mechanism, but we can extract **saliency maps** indicating which pixels most affect the output. Common techniques include gradient-based saliency (calculating gradients of the output with respect to input pixels ⁴⁰), or Class Activation Mapping (CAM) which uses the network's final convolutional features and weights to highlight image regions important for a predicted class. These **XAI (explainable AI) attention maps** for CNNs often show which parts of an image influenced the classification decision ⁴¹. However, multiple studies have found that CNN attention maps differ significantly from human attention: "the similarity between human and CNN attention maps is quite low" ⁴². Humans tend to focus on one or few informative regions (selective, foveal attention), whereas CNN saliency can be more diffuse or scattered ⁴². For example, a CNN diagnosing skin lesions might highlight contextual bits like surrounding skin or artifacts (e.g. fingers holding a lesion) that humans ignore ⁴³. Humans attend to more "discriminative and diagnostic" areas, whereas CNNs sometimes latch onto background context or textures ⁴². Some studies noted CNNs put more weight on context than humans do ⁴². This difference is partly attributed to CNNs being trained for ultimate classification accuracy, not to explicitly mimic human viewing strategy.

That said, CNN architectures and training choices can modulate this alignment. **Deeper networks** and those with more layers tend to have slightly higher human-alignment in some cases ⁴⁴. Networks that incorporate biologically-inspired elements – such as Gaussian receptive field tuning or recurrence – have been shown to attend more like humans ⁴⁴. For instance, using a CNN with human-like receptive field sizes led to a model that focuses more on the primary object (closer to human fixations) ⁴⁵ ⁴⁶. Moreover, the method used to generate the CNN's attention map (e.g. Grad-CAM vs. raw gradients vs. perturbation-based maps) can greatly affect the result ⁴⁷. This variability complicates direct comparison with human saliency, but generally **off-the-shelf CNNs do not naturally "look" at images the way humans do** ⁴² ⁴⁴.

- **Vision Transformers (ViTs):** Transformers bring an explicit attention mechanism to vision models. A ViT divides an image into patches and uses **self-attention** to compute weighted interactions between patches. The learned attention weights can be visualized as an **attention map** between a [CLS] token (representing the image-level class) and the image patches. These maps often highlight important image regions for the Transformer's decision, effectively acting

as saliency maps. ViTs are known to capture global context effectively ⁴⁸, and intriguingly, some self-supervised ViTs have demonstrated emergent object-focused attention heads (e.g. DINO ViT models have heads that outline objects without supervision).

How do ViT attention patterns compare to human saliency? Initial studies suggest mixed results. ViT attention is **more globally diffuse** than human fixations, since the model attends broadly to many patches unless specifically constrained ⁴⁹. Direct correlation between ViT attention and human fixations tends to be low if the model is not guided, similar to CNNs. However, ViTs provide a flexible framework to impose or analyze alignment. Researchers have begun exploring if certain attention heads align with human-like attention. For example, one study found that some ViT heads cluster into patterns, with a subset focusing on objects similarly to human gaze, though others remain dispersed ⁵⁰.

ViTs also enable **adaptive token processing**, which is relevant to saliency. Recent architectures introduce token pruning or adaptive token dropping during inference – essentially skipping patches deemed less important. Algorithms like **DynamicViT** or **Token Adaptive ViT (TAVT)** dynamically remove tokens with low attention weights to save computation ⁵¹ ⁵². This mirrors the idea of a model “foveating” important regions and ignoring unimportant ones, akin to human selective attention. Such adaptive transformers achieve efficiency gains by focusing on salient image content. If the saliency estimation is accurate, these methods can significantly reduce computation with minimal accuracy loss ⁵³. AdaptiveNN can be thought of as an umbrella term for these new networks that **adapt their structure or computation based on input content**, often through learned attention mechanisms. For example, Adaptive Integrated Layered Attention (AILA) networks combine dense skip connections with attention to adapt to image complexity ⁵⁴. Likewise, in CNNs, techniques like Spatial Transformer Networks and glimpse-based recurrent models adaptively focus on sub-regions. An influential example is the **Recurrent Attention Model** (Mnih et al., 2014), which learned to take sequential glances at an image (like saccades) to classify objects, thereby processing only salient parts. These adaptive models draw inspiration from human eye movements and could be ideal for testing human-machine saliency alignment, since they explicitly try to learn where to look.

- **Attention in New Architectures (AdaptiveNN):** Beyond ViTs, other novel architectures incorporate attention for efficiency or robustness. **Adaptive Neural Networks** often use gating or early exiting to allocate computation selectively. For instance, some CNNs can learn to skip certain layers for “easy” images (adaptive depth), or focus high-resolution processing only on likely salient regions (multi-scale foveation). One emerging area is networks that use **foveated vision**: processing a central crop at high detail and periphery at low detail, mimicking human retina. Another is **transformers with adaptive patch sizes**, attending coarsely in unimportant areas and finely where needed. These architectures haven’t been fully explored in terms of how their learned attention maps align with human gaze. It stands to reason that if a machine’s attention allocation strategy is efficient, it might parallel human selective attention to some degree, but this is an open question.

Human–Machine Saliency Alignment and Insights

A central question is whether aligning model attention with human visual saliency has benefits (or drawbacks) for performance and what it reveals about vision. Several comparative studies and benchmarks shed light on how closely machine “gaze” matches human gaze, and the implications:

- **Extent of Alignment:** Research consistently shows that out-of-the-box deep networks do not closely align with human attention during vision tasks ⁴² ⁵⁵. The human–machine attention similarity is usually significantly lower than human–human agreement. One review noted, “human–human similarity is higher than human–CNN similarity” ⁵⁶. Humans looking at the same image tend to agree reasonably on salient regions (especially if a task is defined), whereas a CNN’s saliency map often highlights additional or different areas. As mentioned, CNN attention can be diffuse or include context that humans wouldn’t fixate ⁴². This mismatch has practical ramifications: if an AI model attends to nonsensical image areas (from a human perspective), users may lose trust in the model ⁴¹. This has driven the notion that similarity to human attention is a relevant dimension for evaluating model explanations ⁵⁷ – in other words, we prefer models that focus on “sensible” image regions.
- **Factors Affecting Alignment:** Both machine factors and human factors influence measured alignment. On the machine side, as noted, the choice of architecture and explanation method matters ⁴⁴. Deeper networks and those with human-inspired mechanisms showed higher alignment ⁴⁴. For example, one study found a CNN with neurophysiologically plausible receptive fields (simulating human V1/V2 processing) had attention maps more similar to human fixations ⁴⁴. Another factor is the task and stimuli for humans: alignment can vary if humans are free-viewing versus doing a specific task. **Task instructions** greatly influence human eye movements ⁵⁸ ⁵⁹. If humans are asked to search for an object, their attention might spread broadly or fixate certain cues, whereas a CNN trained for classification always “attends” in service of classification. A recent experiment varied the way human attention maps were collected (eye-tracking vs. mouse-click highlighting vs. segment ranking) and found that the method (free viewing vs. instructed relevance marking) can change the apparent human–CNN similarity ⁶⁰. For instance, when humans manually highlight relevant regions for a given classification, those regions can sometimes correspond better to CNN saliency (since both are tied to the classification decision). Indeed, one finding was that manually selected attention maps were more similar to CNN attention than actual eye fixations in some cases ⁶¹. This indicates that free-viewing fixations include idiosyncrasies (like exploration or viewing bias) that a CNN doesn’t share, whereas if humans are explicitly marking discriminative regions, it aligns more with model logic. **Image complexity** also plays a role: CNNs may do better on simple object-centric images, but in cluttered scenes, human attention often focuses on semantic hotspots (faces, text, etc.), which a purely label-driven model might not consider relevant if those spots aren’t needed for classification.
- **Alignment vs. Performance Trade-off:** A provocative discovery by Fel et al. (2023) was a trade-off between ImageNet classification accuracy and alignment with human visual strategies ⁶². Analyzing 84 CNNs, they found that as networks got more accurate (through scale or advanced architectures), they became **less** aligned with human attention patterns ⁶². In other words, the most accurate models tended to rely on features and strategies quite distinct from human vision. This raises concerns that simply pushing accuracy may lead AI further away from human-like perception. However, Fel et al. did not stop at this observation – they introduced a “neural harmonizer” training procedure that could improve alignment **and** accuracy simultaneously ⁶³. By penalizing discrepancies between model saliency and human saliency during training (using

human attention maps as a form of supervision), they effectively “broke” the trade-off, producing models that were both high-performing and more human-aligned ⁶³ ⁶⁴ . This result is encouraging for the viability of Phase 2: it suggests we can have the best of both worlds with the right training approach. The implication is profound: the scaling laws that led to higher accuracies also led models to use shortcuts or non-human cues, but a targeted alignment loss can steer them to use more human-like features without sacrificing performance ⁶³ .

- **Cognitive and Efficiency Implications:** If a model’s saliency aligns with human gaze, does it mean the model is using more human-like reasoning? Possibly. Humans have evolved efficient strategies to parse scenes – typically focusing on informative regions and ignoring redundancies. A model whose attention map overlaps heavily with human fixations might be prioritizing semantically important features (like objects, faces, text) rather than background texture or random context. There is evidence that models biased towards human-like attention can be more **robust or efficient**. For example, one study found that combining human saliency with machine saliency in biometrics improved recognition performance ⁶⁵ . In iris recognition, filters derived from eye-tracking data (indicating where experts look in iris images) outperformed unguided models ⁶⁵ . For deep networks in object recognition, constraining them to use human-like regions can act as a regularizer against overfitting to peculiar patterns. Another study observed that humans and CNNs often differ in that CNNs are overly sensitive to texture; forcing a model to attend more to shape (which is closer to human strategy) improved its robustness to distortions ⁶⁶ .

On the efficiency front, human attention is highly efficient – we achieve perception by sampling a few key areas. Models that emulate this (like adaptive token Transformers or glimpse models) can drastically cut computation by not processing the whole image at full detail. The WACV 2025 study by Kockwelp et al. demonstrated that using **human gaze to guide a ViT’s attention (via token masking)** allowed the model to use far fewer patches while maintaining or even improving accuracy ⁶⁷ ⁶⁸ . By unmasking only regions where humans fixated, the ViT achieved robust classification with fewer tokens, and in some configurations saw a slight accuracy increase over using the full image ⁶⁸ . This suggests human saliency is an informative prior: the regions humans deem important often **are** those needed for the task, so focusing there is both efficient and effective. However, they also found structural differences remain – meaning even after masking, the pattern of attention inside the model wasn’t identical to humans, indicating that current model attention mechanisms do not perfectly replicate the human process ⁴⁹ ⁶⁹ .

- **Neuroscience (fMRI) Perspective:** An interesting auxiliary route to compare human and machine “attention” is via brain imaging. Recent neuroimaging studies seek to decode saliency or attention from brain activity and compare it to model-derived saliency. For instance, **Calcagno et al. (2026)** used fMRI signals from visual cortex to directly predict spatial saliency maps for viewed images ⁷⁰ . They trained a decoder using DeepGaze II model maps as training targets and showed that the decoded fMRI saliency maps significantly correlate with actual human fixations on images ⁷⁰ ⁷¹ . Notably, they found early visual areas (V1–V4) carried most of the decodable salience information, whereas higher-level regions produced only diffuse, center-biased maps ⁷² ⁷³ . This aligns with the idea that early retinotopic cortex is where low-level saliency is computed, while higher areas modulate attention with task and context (but are harder to decode with fMRI). The ability to reconstruct human attention maps from brain signals opens a door to directly compare neural attention to model attention. If a model’s saliency maps align well with decoded brain-based saliency, it strengthens the case that the model is a good proxy for human internal attention mechanisms. Prior work has also compared deep model layer activations to brain responses (encoding models), finding that models trained on ecologically relevant tasks (or with more human-like architecture) match brain data better ⁷⁴ ⁷⁵ . For example, training CNNs on an

“entry-level” object categorization dataset (Ecoset) rather than fine-grained labels improved the CNN’s similarity to human fMRI patterns ⁴⁶ – presumably because it mirrors the human strategy of categorizing by basic-level categories. These neuroscientific insights, while secondary in our context, support the notion that **human visual attention has identifiable neural signatures** and that aligning models with human attention is measurable at both behavioral (fixation) and neural levels.

Using Human Saliency to Improve Vision Models

Phase 2 of the proposed research explores leveraging human saliency (from eye-tracking or saliency models) as a prior to train or adjust computer vision models. This idea is gaining traction, and several attempts have been made:

- **Regularizing or Guiding Model Training:** One straightforward approach is to add a term in the training loss that encourages the model’s attention (as per an attention map) to match human saliency. For example, the “neural harmonizer” mentioned earlier by Fel et al. did exactly this by aligning model gradients with human attention maps while training on ImageNet ⁶³ ⁶⁴. Another approach, dubbed **CYBORG** (Controlling Your Bias by Overlaying Revealed Gaze), introduced by Sharma et al. (2024), modifies the loss to penalize a model if it puts focus outside human-marked regions ⁷⁶ ⁷⁷. In one of their case studies (iris presentation attack detection), they only changed the loss function (not the architecture) to force the CNN to concentrate on the iris region indicated by human eye-gaze data; as a result, the model’s decision-making became both more accurate and more interpretable to humans (it stopped looking at irrelevant background) ⁷⁸ ⁷⁷. In face recognition and deepfake detection, adding a loss term for human consistency (using human decision patterns or attention) has likewise improved explainability and sometimes accuracy ⁷⁹ ⁸⁰. A notable result: aligning a face recognition CNN with human-like strategies (e.g. focusing on facial features humans use) improved its generalization to edge cases ⁷⁹.
- **Training Data Augmentation with Gaze:** Another line of work uses gaze data to **augment or modify training examples**. The idea is to either mask or blur parts of the image that humans wouldn’t pay attention to, or conversely highlight parts they would. Van Dyck et al. (2022) did experiments guiding a CNN’s attention by altering training images based on human fixations ⁸¹ ⁸². They tried “human-like” training, where images were modified to emphasize human-attended regions, and “non-human-like” training, where images were altered to divert the model’s attention away from what humans focus on. They validated the effect via Grad-CAM saliency maps. The outcome: they successfully trained **non-human-like models** that focused on completely different image parts than humans (serving as a counterfactual), but when attempting to train a **human-aligned model**, they “found no significantly increased human-likeness” from their data augmentation alone ⁸³. In essence, removing everything but the human-fixated regions didn’t automatically make the CNN rely on those regions in the same way humans do. This suggests that while gaze-guided data augmentation can influence model attention, achieving full alignment may require more direct loss constraints or architectural changes. Nonetheless, this approach revealed interesting category-specific effects: for instance, guiding attention away from faces drastically changed the model’s strategy, confirming that face presence heavily influences both human and CNN attention ⁸⁴. It also underscored the role of animacy: images with animals or people had stronger alignment effects, since humans reflexively look at living beings, and CNNs also find them informative for classification ⁸⁴.
- **Human Gaze as a Feature or Input:** Some works have treated human saliency maps as an additional input channel or prior during model training. For example, in an egocentric activity

recognition task, researchers integrated human gaze heatmaps into a spatiotemporal attention module of a deep network, effectively telling the model where humans tend to look in first-person videos ⁸⁵. This improved the recognition accuracy, as the model could pay extra attention to regions likely containing important objects/hands as indicated by human gaze. Similarly, in object detection, one could imagine using human attention maps to bias region proposal networks to look at likely salient locations.

- **Gaze for Token Pruning in Transformers:** The **Human Gaze for ViT Masking** study (Kockwelp et al., 2025) exemplifies using gaze in the inference stage: they masked out patches that humans never looked at and only fed the model patches corresponding to human fixations ⁸⁶ ⁶⁸. This approach yielded two benefits: (1) **Efficiency** – fewer tokens, hence faster and less computation, and (2) sometimes **accuracy gains** – because irrelevant background patches (which might introduce noise or distract the model) were removed, the model made more focused decisions ⁶⁸. Notably, even though the model was not retrained with gaze, this inference-time focus showed that human gaze can serve as a powerful filter. This raises an open question: could we train a ViT (or any model) end-to-end with a mechanism that learns to predict (or use) human-like saliency to decide which parts of the image to process? If the model can learn an internal saliency predictor that matches human distribution, it might achieve big efficiency gains by processing only, say, 30% of the image regions – similar to how we glance at a scene sparsely but still recognize it. Some recent works are moving in this direction by using objectives that reward picking informative patches. Human data could supervise those patch-selection policies.
- **Type of Human Saliency Data Matters:** An important consideration is whether to use **actual eye-tracking data** or **saliency model outputs** (which are predictions of human fixations). High-quality eye-tracking data is ideal but can be hard to collect at scale. If using a computational saliency model as a proxy, the results depend on that model’s accuracy. Some studies have explicitly tested this: e.g. Sharma et al. (CYBORG paper) posed the question “Can human salience be replaced by algorithm-generated salience, or by more training data?” ⁸⁷. Their findings indicated that while using coarse proxy maps (like segmentation masks or classic saliency models) can help, they do not match the benefit of real human saliency ⁸⁷ ⁸⁸. In their experiments, human-provided salience information had a “high value” that wasn’t easily replicated by just feeding the model more data or using generic saliency algorithms ⁸⁸. This suggests that the nuanced differences in where people look – often driven by semantics and human-common sense – carry unique information that improves learning in a way that generic attention maps cannot fully capture. Therefore, one might achieve varying results in Phase 2 depending on the source of saliency prior: using actual eye-tracking from the task domain likely yields the best alignment and performance boost, whereas using a pre-trained saliency prediction model (like DeepGaze) is easier but may introduce its own biases or errors.
- **Interpretability and Trust:** A side benefit of integrating human saliency is enhanced interpretability. If a model is trained to follow human-like attention, its decisions can be easier for humans to interpret – essentially the model points out “I made this decision because I looked here, just like you would.” This has been demonstrated in biometric applications: a post-mortem iris matching tool trained on human-marked salient features became fully interpretable, since it only matched iris patterns that human examiners had identified as important ⁸⁹. In another case, supplying machine-predicted saliency cues to human observers improved the humans’ ability to detect deepfake faces ⁹⁰, hinting at a two-way street: humans can also benefit from machine “attention” guidance if it’s complementary to their own. Aligning the two could create a synergy where humans and AI understand each other’s focus.

Discussion: Viability and Novelty of the Proposed Research

Phase 1 (Human vs. Machine Saliency Comparison) – This review shows that initial work has been done on human–CNN attention comparisons and recently on human–ViT comparisons, but there is plenty of room for comprehensive studies. The proposed systematic comparison across multiple saliency models and multiple vision architectures on standard datasets would be quite novel. Past studies often looked at one model or one type of explanation at a time. For example, most CNN vs human gaze studies used a particular CNN (often VGG or ResNet) and one explanation method (Grad-CAM) ⁹¹ ⁹², sometimes on a specific task like scene classification. A broader benchmark where **saliency maps from: (a) human saliency models (GBVS, DeepGaze II, etc.), (b) CNNs (with Grad-CAM/other), (c) ViTs (self-attention maps), (d) adaptive attention models** are all compared on the same set of images and metrics would provide new insights. We have metrics like NSS and SIM to quantify alignment – using them to rank different vision models by how human-like their attention is would answer questions such as: Do transformers naturally align with human saliency more than CNNs? (some speculate yes, due to global attention, but it needs empirical confirmation), or Do larger models (with higher accuracy) have lower NSS with respect to human fixations? (likely yes, as Fel et al. found a negative correlation ⁶²). It would also be revealing to include newer AdaptiveNN models (like a recurrent attention model or DynamicViT) to see if their learned focus matches human saliency better. If an adaptive model’s goal is efficiency, perhaps it will coincidentally pick out human-salient regions as the most informative parts. This is an open question and an underexplored angle – thus a contribution of Phase 1 could be testing the hypothesis that “models that attend more like humans are more computationally efficient.” We expect some correlation here: models designed to minimize redundancy might converge on salient regions that humans also choose, but it must be validated.

Furthermore, adding an fMRI component in Phase 1 (as a secondary analysis) could be novel. For example, if fMRI-based saliency decoding is available for a dataset ⁷¹, one could check how well each model’s saliency map correlates with the brain-predicted saliency map, not just the eye fixations. This would cross-validate alignment at a neural level. Very few studies, if any, have tried to compare model attention to brain-derived attention signals, so that could be a cutting-edge extension.

Phase 2 (Using Human Saliency in Training) – The literature confirms this is a promising but challenging direction. Several pioneering works have tried incorporating human attention, with mixed results: some show improved performance and alignment ⁶³ ⁶⁵, others found it non-trivial to increase human-likeness without complex changes ⁸¹ ⁹³. This means there is still much to explore, and the proposed research can contribute significantly. Potential novel components include:

- Trying different types of saliency priors (actual eye-tracking heatmaps vs. high-quality model-generated saliency) as training signals and analyzing which works best. For example, using the SALICON model’s output as a cheaper prior and comparing it to using real eye-tracking from MIT1003 could reveal how important genuine human data is.
- Investigating **selective attention training** for efficiency: e.g., train a network with an auxiliary loss to minimize computation on non-salient regions (perhaps by blurring or masking them during training based on human maps). This goes beyond just matching saliency – it directly uses human saliency to prune the model’s processing. No work has yet definitively shown that human gaze can cut down FLOPs in a controlled training experiment, so a study that trains a “foveated CNN” using human gaze as a guide could be very novel.
- Exploring **task generalization**: Does a model trained with human saliency prior on one task (say classification) also show benefits on another (detection or segmentation)? If human attention truly captures generally important regions, it might help across tasks.

- Examining **which human saliency model is best as a prior**: GBVS vs. DeepGaze vs. SALICON – these differ (GBVS is more low-level; DeepGaze is very high-level with deep features). The transferability might vary: perhaps deep learned saliency (which contains semantic knowledge) is more useful to guide a ResNet, whereas classic saliency might not provide anything beyond what the CNN already computes in early layers. This is relatively unexplored, and evaluating a few could be insightful.

In terms of novelty, Phase 2 touches on a contemporary frontier: aligning AI with human cognitive patterns. Fel et al. (2023) is one of the first to show performance can improve with alignment ⁶³, but their method was for object recognition only. There is still much unknown about why and when human-like attention helps. Could it ever hurt performance? Possibly if human gaze includes task-irrelevant fixations (e.g. humans sometimes look at a person’s face in an image even if the task is to classify the scene – a model might not need that for the label). The proposal to test whether human-like saliency improves **generalization** is compelling: if a model learns to rely on broadly applicable salient cues (like objects) rather than dataset-specific backgrounds, it should generalize better to new data. Some evidence for this: models trained to be human-aligned (harmonized models) were reported to avoid the texture bias and reliance on background that plague standard CNNs ⁶⁶ ⁹⁴. Thus, they might be more robust to distribution shifts. This remains underexplored and would be a valuable contribution.

Challenges and Considerations: Collecting high-quality human attention data for training can be a bottleneck. However, large datasets like SALICON and newly released ones with eye-tracking for COCO images exist, providing at least a starting point. Ensuring that the metrics (NSS, etc.) guide training in the right direction is another challenge – multi-object images and multiple humans have stochastic attention, so there’s noise in ground truth. The research must also consider that human attention is not monolithic; it varies by observer and by task. So “human saliency” used for training should match the context of the model’s task (free-viewing data may not perfectly suit a fine-grained classification task, for example). A novel component could be dynamically weighting saliency loss: e.g., only enforce alignment on certain “key” images or regions that are known to be important, to avoid forcing the model to pay attention to genuinely irrelevant but eye-catching details (like a colorful background that humans gaze at briefly but isn’t needed for recognition).

In summary, the proposed research direction is well-grounded in current literature and addresses gaps at the intersection of vision science and AI. **Phase 1** will bring a unifying comparative analysis that can reveal trends (e.g., transformers vs CNN in saliency alignment) and test assumptions (like alignment vs efficiency). **Phase 2** will push into relatively unexplored territory of using human cognitive data to directly shape AI training; early works show promise, but many questions remain, making this phase highly timely. The novelty lies in combining these aspects: few works have attempted to use human saliency systematically to improve models, and none have closed the loop by evaluating if those improvements feed back into more human-like attention in a virtuous cycle.

Conclusion

There is growing evidence that bridging human and machine visual saliency is a fruitful endeavor. Human saliency prediction models are now very advanced, providing a solid basis for comparison and training data. Machine vision models, from CNNs to ViTs and adaptive networks, offer diverse mechanisms of attention that we can analyze and influence. Aligning a model’s attention with that of humans may lead to models that not only perform better and more efficiently, but are also more interpretable and cognitively plausible. This literature review highlights that while humans and current AI often diverge in “where they look,” we now have the tools (datasets, metrics, and initial algorithms) to quantify and reduce that divergence. The proposed two-phase research would extend the state of the art by thoroughly comparing saliency across systems and by pioneering new ways to inject human-like

attention into AI. Such research is poised to deepen our understanding of vision, both biological and artificial, and pave the way for AI that sees the world a bit more like we do. It is an exciting, interdisciplinary venture with implications for cognitive science, AI efficiency, and the development of human-aligned intelligent systems.

Sources: The information and examples above were drawn from a range of recent studies, including reviews of saliency models ¹⁵ ¹⁶, analyses of human vs CNN attention ⁴² ⁴⁴, transformer attention research ⁶⁸, and novel human-in-the-loop training approaches ⁶⁵ ⁶², among others, as cited throughout this report.

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